

Matheus Augusto de P. Oliveira

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Summary

Python Developer specialized in Computer Vision and Biometrics, focused on image processing and Machine Learning pipelines. CNPq Research Fellow at UTFPR developing desktop tools for neonatal biometric analysis in partnership with Akiyama Group/Infantid.

Education

2021/2 - Present	Computer Engineering – Federal University of Technology – Paraná (UTFPR), Pato Branco (expected graduation: 2026/1)
2018 - 2020	Technical Diploma in Industrial Automation and Mechatronics – ETEC Martin Luther King, São Paulo

Experience

CNPq Undergraduate Research Scholar	2024 – Present
<i>UTFPR / Akiyama Group – Biometrics and Computer Vision</i>	

- Developed three Python tools (BenaPRO, Bena, and supporting scripts) for a biometrics pipeline: multi-frame alignment, multi-label annotation with severity scoring, and automated validation on neonatal fingerprint datasets.
- Implemented preprocessing routines (NumPy, OpenCV) and validation workflows comparing manual vs. automated results, exporting structured outputs (JSON/CSV) for reproducibility and model training.
- Curated datasets, technical documentation, and version-controlled experiments (Git), supporting research decisions on isotropic growth and biometric uniqueness.

Projects

- **BenaPRO – Dataset Annotation Tool** *Python, PyQt6, OpenCV*
Developed a desktop application to classify fingerprint capture errors (multi-label). Focused on UX to speed up expert work and ensure consistency in training data.
- **Bena – Fingerprint Alignment Tool** *Python, OpenCV, NumPy*
GUI software for assisted manual alignment across multiple frames and captures taken at different times. Exports a complete package with processed images and a structured log of applied transformations.
- **M.T.E.X – Mobile Robotics for Science Outreach** *C++, ESP32, Arduino*
End-to-end prototyping of an award-recognized mobile robot (Honorable Mention). Integrated distance sensors, motor drivers, and autonomous control logic for basic navigation and obstacle detection.
- **L.I.T.S – Brazilian Sign Language (Libras) Translator Glove** *C++, Arduino, Flex Sensors*
Wearable device with five flex sensors capturing Libras gestures and converting them to synthesized audio (DFPlayer Mini). Recognizes the fingerspelling alphabet and 50+ common everyday words.

Skills

- **Programming languages:** Python (advanced), C/C++, SQL, MATLAB/Octave (intermediate), Kotlin, Lua, Java, JavaScript/HTML/CSS (basic).
- **Libraries:** OpenCV, NumPy, Pandas, PyQt5/6, Matplotlib, Tkinter, scikit-learn, dlib.
- **Tools:** Git/GitHub, LaTeX, Linux (Ubuntu), Arduino IDE, ESP-IDF, PlatformIO, VS Code, Proteus, CoppeliaSim, MATLAB/Simulink, Jupyter Notebook.
- **Domains:** Computer vision, embedded systems, biometrics, data engineering (ETL, pipelines, validation), tool development (CLI/GUI), software engineering (version control, testing, documentation), product prototyping, mobile robotics, IoT.
- **Hardware:** Microcontrollers (Arduino, ESP32), SBCs (Raspberry Pi), sensors (ultrasonic, flex, PIR, LiDAR, IMU, encoders), actuators (DC/servo/omnidirectional motors), power drivers, audio modules, cameras, power supplies and batteries.

Publications

- Isotropic growth analysis in newborn fingerprints (Portuguese title). SEI/SICITE UTFPR, 2025.
- M.T.E.X: autonomous mobile robotics for science outreach (Portuguese title). UTFPR Tech Week, 2025.

Languages

Portuguese: Native | **English:** Fluent technical reading, intermediate conversation | **Spanish:** Basic